PROJECT DESIGN PHASE

Problem - Solution Fit Template

Date	27 September 2026
Team ID	23KD1A1507
Project Name	Automated Network Request Management Using ServiceNow
Maximum Marks	2 Marks

Problem - Solution Fit Template:
The Problem-Solution Fit means that a real problem has been identified for the customer and that the solution actually solves it. The canvas below applies it to the Automated Network Request Management System.
Purpose / Vision: Replace repetitive manual ticket booking, network request forms, approval chains, and status updates with a centralized, automated ServiceNow application.

1. CUSTOMER SEGMENT(S) [CS]

- Enterprise employees needing network service requests (IP allocation, firewall ports, VPN).
- IT support desk staff and system administrators who process and manage network requests.
- Network engineering teams and approvers responsible for validating and fulfilling infrastructure access.

6. CUSTOMER CONSTRAINTS [CC]

- Limited time to wait for manual email approvals and port provisioning.
- Uncertainty about required network parameters and request types.
- Need for a simple process that does not require repeated entry of details.
- Prototype limits: simulated network switch orchestration, simplified approval workflows.

5. AVAILABLE SOLUTIONS [AS]

• Manual email requests and physical ticket forms.

Pros: Familiar; IT staff can handle special cases in person.

Cons: Long wait times, manual data entry, risk of configuration errors, records spread across places, manual status updates.

2. JOBS-TO-BE-DONE / PROBLEMS [J&P]

• **Requester:** choose connection type, know required parameters, get a ticket with a Request ID and confirmation.

• **Staff:** record requests accurately, keep network table data up to date, track request status.

Problem: these jobs rely on repetitive manual steps.

9. PROBLEM ROOT CAUSE [RC]

- Approval tracking depends on manual work or separate email threads.
- Request records and status are not managed in one centralized place.
- No automated rule or workflow to validate inputs, identify tickets, and update status consistently.

7. BEHAVIOUR [BE]

- Requesters send emails to IT, ask for status, and wait to be served.
- Staff enter request details by hand and update the request status manually.
- Staff look in different spreadsheets/inboxes to find request information.

3. TRIGGERS [TR]

- Need to access restricted network resources or ports.
- Experiencing long delays or being unsure of request status.
- A new request being submitted (starts flow and status routing).

4. EMOTIONS: BEFORE/AFTER [EM]

Before: rushed, uncertain about request approval, frustrated by waiting.

After: confident, in control, assured that the request is confirmed and identifiable by its ticket number.

10. YOUR SOLUTION [SL]

A ServiceNow application with Service Catalog, Custom Database Table (`u_database_table`), and Tasks tables.

- Catalog Client Scripts & UI Policies validate parameters dynamically.
- Each request has a unique ticket auto-number (`Database Number`).
- Flow Designer flow "Network Request Automation" sets status to Confirmed/Approved on record creation.
- System Email Logs prepared for automated email notifications.

8.1 ONLINE [CH]

Digital request records and catalog forms in ServiceNow Service Portal (`/sp`).

• Future: automated provisioning integration, email alerts, real-time dashboard widgets.

8.2 OFFLINE [CH]

• IT helpdesk staff and network admins.
• In-person enquiries and paper-based or manual records.

Note: This is a prototype. Hardware switch auto-provisioning, notifications, external gateway integrations, and self-service analytics interfaces are proposed extensions, not completed features.

References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe